

PH101 (II) Tutorials

Tutorial 1, 2, 3

Updated on October 23, 2023

Contents

1	Tutorial 1	2
1.1	Problems	2
1.2	Solutions	3
2	Tutorial 2	9
2.1	Problems	9
2.2	Solutions	11
3	Tutorial 3	15
3.1	Problems	15
3.2	Solutions	16

1 | Tutorial 1

Click on [Problem x.](#) to go to the solution to problem x.

1.1 • Problems

Problem 1. Show that the wave equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0$$

is not invariant under Galilean transformation, where c is the speed of light.

Problem 2. A train with proper length L moves at a speed $5c/13$ with respect to the ground. A ball is thrown from the back of the train to front. The speed of the ball with respect to the train is $c/3$. As viewed by someone on the ground, how much time does the ball spend in the air, and how far does it travel?

Problem 3. Two events **A** and **B** are described by space-time coordinates in **S** frame by

A: $x_1 = 10$ km, $y_1 = 0$, $z_1 = 0$, $t_1 = 2 \times 10^{-5}$ sec;

B: $x_2 = 100$ km, $y_2 = 0$, $z_2 = 0$, $t_2 = 10^{-5}$ sec.

Find out if there is an inertial frame **S'** (moving with velocity v along the $x = x'$ axis) in which **A** and **B** are simultaneous. If so, find its velocity with respect to **S**.

Problem 4. Triplet A stays on the earth. Triplet B travels at speed $4c/5$ to a planet (a distance L away from earth) and back. Triplet C travels out to the planet at speed $3c/4$, and then returns at the necessary speed to arrive back exactly when B does. [Triplets are of same age by birth.]

(a) How much does each triplet age during this process?

(b) Who is youngest?

Problem 5. A spaceship is moving away from the earth at a speed $v = 0.8c$. When the spaceship is at a distance of 6.66×10^8 km from the earth as measured in earth's reference frame, a radio signal is sent out to the spaceship by an observer on the earth. How long will it take for the signal to reach the spaceship

(a) as measured in the earth's reference frame.

(b) as measured in spaceship's reference frame.

Problem 6. A train and a tunnel both have proper lengths L . The train speeds toward the tunnel, with speed v . A bomb is located at the front of the train. The bomb is designed to explode when the front of the train passes the far end of the tunnel. A deactivation sensor is located at the back of the train. When the back of the train passes the near end of the tunnel, this sensor *tells* the bomb to disarm itself. Does the bomb explode?

1.2 • Solutions

Some possible solutions for the problems of Tutorial 1. Click on [Solution to Problem x.](#) to go back to problem x.

Solution to Problem 1.

Show that the wave equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0$$

is not invariant under Galilean transformation, where c is the speed of light.

The wave function is given by $\phi(x, y, z, t)$. Under Galilean transformation:

- $x' = x - vt \implies \partial x / \partial x' = 1, \partial y / \partial x' = \partial z / \partial x' = \partial t / \partial x' = 0$. By the Chain Rule,

$$\begin{aligned} \frac{\partial \phi}{\partial x'} &= \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial x'} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial x'} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial x'} + \frac{\partial \phi}{\partial t} \cdot \frac{\partial t}{\partial x'} \\ &= \frac{\partial \phi}{\partial x} \cdot 1 + 0 + 0 + 0 \\ \implies \frac{\partial \phi}{\partial x'} &= \frac{\partial \phi}{\partial x}. \end{aligned}$$

Note that the above steps hold for any arbitrary function of (x, y, z, t) . Partially differentiating again with respect to x' , we obtain:

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x'^2} &= \frac{\partial}{\partial x'} \frac{\partial \phi}{\partial x} \\ &= \frac{\partial}{\partial x'} \psi \\ &= \frac{\partial}{\partial x} \psi \\ \implies \frac{\partial^2 \phi}{\partial x'^2} &= \frac{\partial^2 \phi}{\partial x^2} \end{aligned}$$

- $y' = y, z' = z$. Clearly,

$$\frac{\partial^2 \phi}{\partial y'^2} = \frac{\partial^2 \phi}{\partial y^2} \quad \text{and} \quad \frac{\partial^2 \phi}{\partial z'^2} = \frac{\partial^2 \phi}{\partial z^2}.$$

- $t' = t$. $x' = x - vt = x - vt' \implies x = x' + vt' \implies \partial x / \partial t' = v, \partial y / \partial t' = \partial z / \partial t' = 0, \partial t / \partial t' = 1$. Using Chain Rule, we have

$$\begin{aligned} \frac{\partial \phi}{\partial t'} &= \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial t'} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial t'} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial t'} + \frac{\partial \phi}{\partial t} \cdot \frac{\partial t}{\partial t'} \\ &= \frac{\partial \phi}{\partial x} \cdot v + 0 + 0 + \frac{\partial \phi}{\partial t} \\ \implies \frac{\partial \phi}{\partial t'} &= v \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial t}. \end{aligned}$$

Differentiating again with respect to t' ,

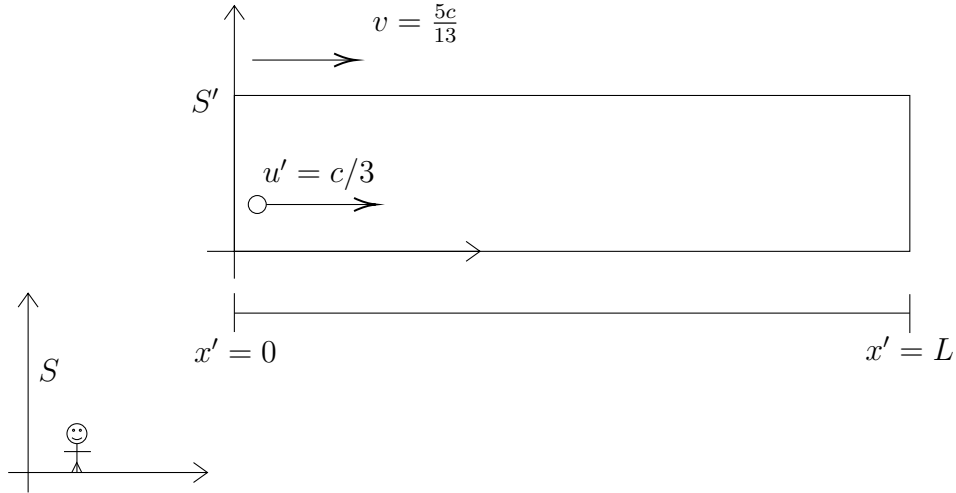
$$\begin{aligned}
 \frac{\partial^2 \phi}{\partial t'^2} &= \frac{\partial}{\partial t'} \left(v \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial t} \right) \\
 &= v \frac{\partial}{\partial t'} \frac{\partial \phi}{\partial x} + \frac{\partial}{\partial t'} \frac{\partial \phi}{\partial t} \\
 &= v \frac{\partial}{\partial t'} \psi_1 + \frac{\partial}{\partial t'} \psi_2 \\
 &= v \left(v \frac{\partial \psi_1}{\partial x} + \frac{\partial \psi_1}{\partial t} \right) + \left(v \frac{\partial \psi_2}{\partial x} + \frac{\partial \psi_2}{\partial t} \right) \\
 \implies \frac{\partial^2 \phi}{\partial t'^2} &= v^2 \frac{\partial^2 \phi}{\partial x^2} + 2v \frac{\partial^2 \phi}{\partial x \partial t} + \frac{\partial^2 \phi}{\partial t^2} \neq \frac{\partial^2 \phi}{\partial t^2},
 \end{aligned}$$

and hence, the wave equation is not invariant under Galilean transformation.

Solution to Problem 2.

A train with proper length L moves at a speed $5c/13$ with respect to the ground. A ball is thrown from the back of the train to front. The speed of the ball with respect to the train is $c/3$. As viewed by someone on the ground, how much time does the ball spend in the air, and how far does it travel?

Refer to the figure. Let the train be the S' -frame with $v = 5c/13$. Then, $\gamma = \frac{1}{\sqrt{1-v^2/c^2}} = 13/12$.



The time taken by the stone in the frame of the train, t' is given by

$$t' = \frac{L}{c/3} = \frac{3L}{c}.$$

By Inverse Lorentz Transformation with $t'_i = 0, t'_f = t' = 3L/c$ and $x'_i = 0, x'_f = L$ we obtain

$$\begin{aligned}
 \Delta x &= x_f - x_i \\
 &= \gamma(x'_f + vt'_f) - \gamma(x'_i + vt'_i) \\
 &= \frac{13}{12} \left(L + \frac{5c}{13} \cdot \frac{3L}{c} \right) = \frac{7L}{3}, \quad \text{and} \\
 \Delta t &= t_f - t_i \\
 &= \gamma(t'_f + vx'_f/c^2) - \gamma(t'_i + vx'_i/c^2) \\
 &= \frac{13}{12} \left(\frac{3L}{c} + \frac{5c}{13} \cdot \frac{1}{c^2} \cdot L \right) = \frac{11L}{3c}.
 \end{aligned}$$

Solution to Problem 3.

Two events **A** and **B** are described by space-time coordinates in **S** frame by

A: $x_1 = 10 \text{ km}, y_1 = 0, z_1 = 0, t_1 = 2 \times 10^{-5} \text{ sec};$

B: $x_2 = 100 \text{ km}, y_2 = 0, z_2 = 0, t_2 = 10^{-5} \text{ sec}.$

Find out if there is an inertial frame **S'** (moving with velocity v along the $x = x'$ axis) in which **A** and **B** are simultaneous. If so, find its velocity with respect to **S**.

For simultaneous events, $t'_1 = t'_2$ in the frame S' . So,

$$\begin{aligned}
 0 &= t'_2 - t'_1 \\
 &= \gamma(t_2 - \frac{v}{c^2}x_2) - \gamma(t_1 - \frac{v}{c^2}x_1) \\
 &= (t_2 - t_1) - \frac{v}{c^2}(x_2 - x_1) \\
 &= -10^{-5} - \frac{v}{c^2} \cdot (9 \times 10^4) \\
 \implies v &= -10^7 \text{ m/sec.}
 \end{aligned}$$

Solution to Problem 4.

Triplet A stays on the earth. Triplet B travels at speed $4c/5$ to a planet (a distance L away from earth) and back. Triplet C travels out to the planet at speed $3c/4$, and then returns at the necessary speed to arrive back exactly when B does. [Triplets are of same age by birth.]

- How much does each triplet age during this process?
- Who is youngest?

Easy. Algorithmic. Fantastic. Fix the S frame as the Earth with observer A. The return of both triplets B and C is simultaneous according to A. We assume that both B and C start their respective journeys simultaneously at $t = 0$ according to A.

The following steps should be followed:

- **Step 1.** Define the event: *Triplet B returns to Earth.* We wish to compute the time taken by B. Note that, **B has to cover distance $2L$ with a speed $4c/5$ in Earth's frame.** Hence, the time taken by B is $2L/(4c/5) = 5L/2c$. This is the time spent, according to A. Call it t_A . So, $t_A = 5L/2c = 2.5L/c$.
- **Step 2.** Note that, the final position of B is same as initial position (in S-frame). Let $\gamma_B = 1/\sqrt{1 - (4c/5)^2/c^2} = 5/3$. So, by the time dilation result, the time spent in B's frame, say t_B , will be $t_B = t_A/\gamma_B = 3L/2c = 1.5L/c$.
- **Step 3.** What about C? Define the event: *Triplet C reaches the planet.* **C covers distance L with speed $3c/4$ in Earth's frame.** The time t_{C_1} spent in Earth's frame equals $L/(3c/4) = 4L/3c$. Now define the return journey as another event: *Triplet C returns from the planet to Earth.* Let C takes time t_{C_2} for this journey. As B and C return at the same time, we know $t_A = t_{C_1} + t_{C_2} \implies t_{C_2} = 7L/6c = L/(6c/7)$. So, the speed of C in the return journey is $6c/7$. Let $\gamma_{C_1} = 1/\sqrt{1 - (3c/4)^2/c^2} = 4/\sqrt{7}$ and $\gamma_{C_2} = 1/\sqrt{1 - (6c/7)^2/c^2} = 7/\sqrt{13}$. Using the results of Time dilation, the time spent in C's frame,

$$t_C = \frac{t_{C_1}}{\gamma_{C_1}} + \frac{t_{C_2}}{\gamma_{C_2}} = \left(\frac{2\sqrt{7} + \sqrt{13}}{6} \right) \frac{L}{c} \approx 1.48L/c.$$

Hence, the correct order of ages is $C < B < A$.

Solution to Problem 5.

A spaceship is moving away from the earth at a speed $v = 0.8c$. When the spaceship is at a distance of 6.66×10^8 km from the earth as measured in earth's reference frame, a radio signal is sent out to the spaceship by an observer on the earth. How long will it take for the signal to reach the spaceship

- (a) as measured in the earth's reference frame.
- (b) as measured in spaceship's reference frame.

The radio signal is essentially a light pulse with speed c .

- (a) Let $L = 6.66 \times 10^{11}$ m and time taken be t . Then, $ct = vt + L \implies t = \frac{L}{c-v} = 1.11 \times 10^4$ sec.
- (b) **WE CANNOT USE TIME DILATION HERE as the time interval is not measured at the same place.** Then, what to do? Simple. Lorentz.

Consider the events —

- E_0 : signal sent by observer on earth,
- E_1 : signal arrives at spaceship.

Well, it is easy to see that $E_0 = (0, 0, 0, t_0)$ and $E_1 = (ct, 0, 0, t_0 + t)$. Let $t_1 = t_0 + t \implies t' = t'_1 - t'_0$. By Lorentz Transformation, and using $\gamma = 1/\sqrt{1 - (0.8c)^2/c^2} = 5/3$,

$$\begin{aligned} t' &= t'_1 - t'_0 \\ &= \gamma \left(t_1 - \frac{0.8c}{c^2} \cdot x_1 \right) - \gamma \left(t_0 - \frac{0.8c}{c^2} \cdot x_0 \right) \\ &= \gamma t - \frac{0.8}{c} \gamma \cdot (ct) \\ &= 0.2\gamma t = 3.7 \times 10^3 \text{ sec} \end{aligned}$$

where $x_1 = ct$ and $x_0 = 0$ denote the x -coordinates of the events E_1 and E_0 respectively in the Earth's frame.

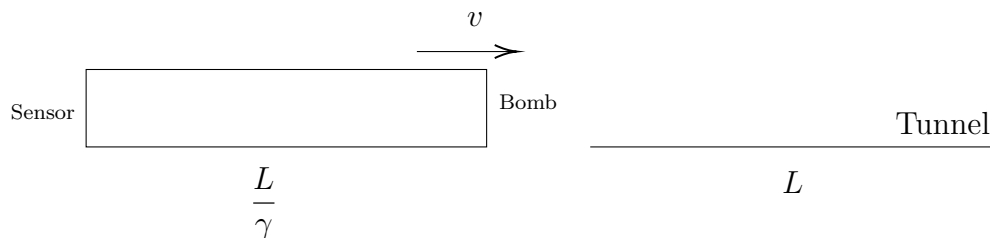
Solution to Problem 6.

A train and a tunnel both have proper lengths L . The train speeds toward the tunnel, with speed v . A bomb is located at the front of the train. The bomb is designed to explode when the front of the train passes the far end of the tunnel. A deactivation sensor is located at the back of the train. When the back of the train passes the near end of the tunnel, this sensor *tells* the bomb to disarm itself. Does the bomb explode?

The Train-tunnel paradox! We present two approaches, (1) from the train frame and (2) the tunnel frame. *One important aspect of testing conceptual clarity in relativity is the ability to successfully cross-check the consistency of the answers from various reference frames.*

Let $\gamma = 1/\sqrt{1 - v^2/c^2} > 1$.

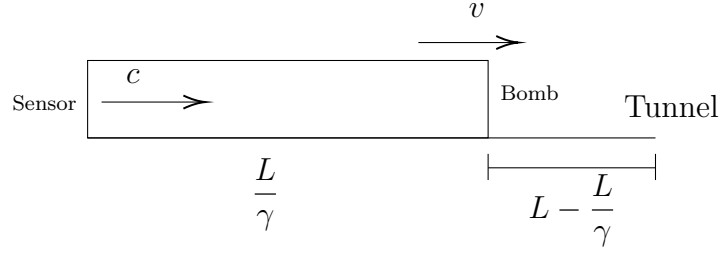
- **Approach 1 (Train frame).** The length of the train is L . The tunnel, contracted to length $L/\gamma < L$ approaches it with speed v . So, the tunnel crosses the bomb first (equivalently, the bomb crosses the tunnel first) and hence the bomb blasts.
- **Approach 2 (Tunnel frame).** The length of the train gets contracted to L/γ . The



situation is as in the figure.

Easily, the sensor reaches the tunnel before the bomb crosses it. *So, it seems that the bomb should not blast! But, the bomb clearly blasts in the train's frame and so, it must blast from every other inertial frame as well!* Where's the problem, then? *Quite tricky, actually.* Well, it will take some time for the sensor to *tell* the bomb to disarm itself.

We now wish to calculate the time taken by a signal — the best case, a light pulse — to travel from the sensor to the bomb. We present two ways to calculate it:



- *Method 1:* From the figure, the time taken by the pulse from the sensor equals $\frac{L/\gamma}{c - v}$.
- *Method 2 (Lorentz Transformation):* Note that the time taken by the pulse in the train frame equals $t' = L/c$. Consider the sensor at $x' = 0$ and bomb at $x' = L$ in the train frame. Then, the time taken, t , in the tunnel frame equals (by Inverse Lorentz Transformation)

$$\begin{aligned}
 t &= \gamma \left(t' + \frac{v}{c^2} x' \right) \\
 &= \gamma \left(\frac{L}{c} + \frac{v}{c^2} L \right) \\
 &= \frac{\gamma L}{c} \left(1 + \frac{v}{c} \right) = \frac{L/\gamma}{c} \cdot \left(\gamma^2 \left(1 + \frac{v}{c} \right) \right) = \frac{L/\gamma}{c} \left(\frac{1}{1 - \frac{v}{c}} \right) = \frac{L/\gamma}{c - v}.
 \end{aligned}$$

Now, the time taken by the bomb to cross the tunnel equals $\frac{L - L/\gamma}{v}$. If the bomb is not to explode, then the time taken by the signal must be lesser than the time of crossing the tunnel which means

$$\begin{aligned}
 \frac{L/\gamma}{c - v} &< \frac{L - L/\gamma}{v} &\iff \frac{1}{c - v} + \frac{1}{v} &< \frac{\gamma}{v} \\
 &&\iff \frac{c}{v(c - v)} &< \frac{\gamma}{v} \\
 &&\iff \frac{c}{c - v} &< \gamma \\
 &&\iff \frac{1}{\sqrt{1 - \frac{v}{c}}} &< \frac{1}{\sqrt{1 + \frac{v}{c}}} \\
 &&\iff \sqrt{1 + \frac{v}{c}} &< \sqrt{1 - \frac{v}{c}},
 \end{aligned}$$

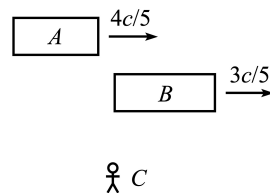
which is absurd. Hence, the signal cannot reach before the bomb and so, the bomb will always explode.

2 | Tutorial 2

Click on [Problem x.](#) to go to the solution to problem x.

2.1 • Problems

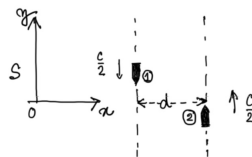
Problem 1. Two trains, **A** and **B**, each have proper length L and move in the same direction. **A**'s speed is $4c/5$, and **B**'s speed is $3c/5$ with respect to ground. **A** starts behind **B** (see figure).



- How long, as viewed by a person **C** on the ground, does it take for **A** to overtake **B**?
- At what speed, **B** would see **A** to cross?
- Explain the phenomena of **A** crossing **B** from the reference frame of **A**.

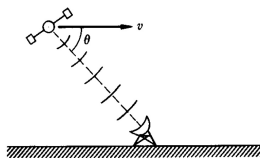
[By the *time of overtaking*, we mean time interval between the front of **A** passing the back of **B**, and the back of **A** passing the front of **B**.]

Problem 2. As observed in an inertial frame S , two spaceships are travelling in opposite directions along straight, parallel trajectories separated by a distance d as shown in the figure. The speed of each ship is $c/2$.



- At the instant when the ships are at the points of closest approach, ship-1 ejects a small package which has speed $3c/4$ as viewed from S . From the point of view of the observer in ship-1, at what angle must the package be aimed for it to be received by ship-2?
- What is the speed of the package as seen by the observer in ship-1?

Problem 3. Consider a Satellite which broadcasts a radio beacon signal (continuous signal), is moving with a velocity v at an angle θ with respect to a ground tracking station (see the figure below), where the angle θ is measured in the rest frame of the observer. Find the frequency ν_D observed by the station if the frequency of the signal emitted is ν_0 from rest frame of the Satellite.



Problem 4. The kinetic energy (K) and the momentum (p) of a particle deduced from measurements on its track in nuclear experiment are 250 MeV and 368 MeV/c, respectively. Determine the rest mass of the particle.

Problem 5. A π meson decays into a muon (μ) and a neutrino (ν_μ) via the following reaction. The neutrino has zero rest mass.

$$\pi^+ \rightarrow \mu^+ \nu_\mu.$$

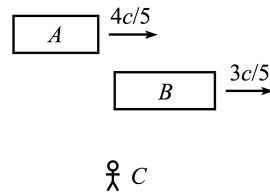
- (a) Using the relationship between total relativistic energy and momentum, show that the magnitude of the momentum p of the decay products in the pion rest frame is given by $p/c = (M_0^2 - m_0^2)/2M_0$; where M_0 is the pion rest mass and m_0 is the muon rest mass.
- (b) Consider the decay life time to be 2.5×10^{-8} sec in the rest frame of pion. Assuming that the π meson has kinetic energy equal to its rest energy, what distance would the meson travel before decaying as seen by an observer at rest?

2.2 • Solutions

Click on [Solution to Problem x.](#) to go back to problem x.

Solution to Problem 1.

Two trains, **A** and **B**, each have proper length L and move in the same direction. **A**'s speed is $4c/5$, and **B**'s speed is $3c/5$ with respect to ground. **A** starts behind **B** (see figure).



- How long, as viewed by **C** on the ground, does it take for **A** to overtake **B**?
- At what speed, **B** would see **A** to cross?
- Explain the phenomena of **A** crossing **B** from the reference frame of **A**.

[By the *time of overtaking*, we mean time interval between the front of **A** passing the back of **B**, and the back of **A** passing the front of **B**.]

(a) Time taken (in ground frame) = $\frac{\frac{L}{\gamma_A} + \frac{L}{\gamma_B}}{4c/5 - 3c/5} = 7L/c$.

- (b) Using Velocity addition,

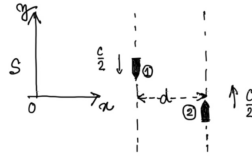
$$\frac{4c/5 - 3c/5}{1 - \frac{1}{c^2} \cdot \frac{4c}{5} \cdot \frac{3c}{5}} = 5c/13.$$

- (c) In frame of **A**, **B** appears to be approaching with speed $5c/13$ and a contracted length $L\sqrt{1 - (5c/13)^2/c^2} = 12L/13$. In order to overtake, distance to be covered equals $L + 12L/13 = 25L/13$. So, time taken in frame of **A** equals $(25L/13)/(5c/13) = 5L/c$.

Solution to Problem 2.

As observed in an inertial frame S , two spaceships are travelling in opposite directions along straight, parallel trajectories separated by a distance d as shown in the figure. The speed of each ship is $c/2$.

- At the instant when the ships are at the points of closest approach, ship-1 ejects a small package which has speed $3c/4$ as viewed from S . From the point of view of the observer in ship-1, at what angle must the package be aimed for it to be received by ship-2?



(ii) What is the speed of the package as seen by the observer in ship-1?

Let velocity of package be $u_x\mathbf{i} + u_y\mathbf{j}$. Package reaches ship-2. So, $u_y = c/2$. We know $3c/4 = \sqrt{u_x^2 + u_y^2} \implies u_x = \sqrt{5}c/4$. Let $\mathbf{u}' = u'_x\mathbf{i} + u'_y\mathbf{j}$ be the velocity of the package with respect to ship-1. By Velocity Addition,

$$\begin{aligned} u'_x &= \frac{u_x/\gamma}{1 - u_x v/c^2} \\ &= \frac{\sqrt{5}c/4 \cdot \sqrt{3}/2}{5} \\ &= \frac{\sqrt{15}c}{10}, \end{aligned}$$

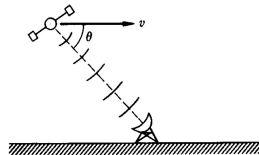
$$\begin{aligned} u'_y &= \frac{u_y - (-c/2)}{1 - u_y \cdot (-c/2)/c^2} \\ &= \frac{c/2 - (-c/2)}{1 + 1/4} \\ &= 4c/5. \end{aligned}$$

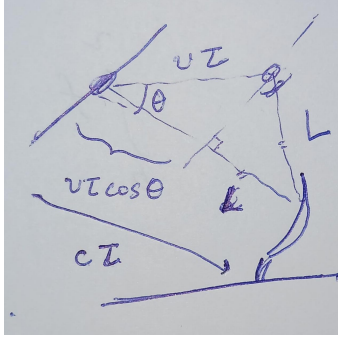
(i) Angle, with the ship 1 (y-axis) = $\arctan(u'_x/u'_y) = \arctan(\sqrt{15}/8) \approx 25.83^\circ$ with the $+y$ -axis.

(ii) Speed equals $\sqrt{u'^2_x + u'^2_y} = \sqrt{79}c/10 \approx 0.89c$.

Solution to Problem 3.

Consider a Satellite which broadcasts a radio beacon signal (continuous signal), is moving with a velocity v at an angle θ with respect to a ground tracking station (see the figure beside), where the angle θ is measured in the rest frame of the observer. Find the frequency ν_D observed by the station if the frequency of the signal emitted is ν_0 from the rest frame of the Satellite.





Refer to the figure. Assume the signal is emitted as a pulse. Then, we have

$$c\tau = L + v\tau \cos \theta.$$

For the station,

$$\begin{aligned} \nu_D &= \frac{c}{L} \\ &= \frac{c}{(c - v \cos \theta)(\tau)} \\ &= \frac{c}{(c - v \cos \theta)(\gamma\tau_0)} \\ &= \nu_0 \frac{\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c} \cos \theta}. \end{aligned}$$

Solution to Problem 4.

The kinetic energy (K) and the momentum (p) of a particle deduced from measurements on its track in nuclear experiment are 250 MeV and 368 MeV/c, respectively. Determine the rest mass of the particle.

$$\begin{aligned} E^2 &= m_0^2 c^4 + p^2 c^2 \\ \implies (K + m_0 c^2)^2 &= m_0^2 c^4 + p^2 c^2 \\ \implies 2m_0 K c^2 &= p^2 c^2 - K^2 \\ \implies m_0 &= \frac{p^2}{2K} - \frac{K}{2c^2} \approx 145.848 \text{ MeV}/c^2 \end{aligned}$$

Solution to Problem 5.

A π meson decays into a muon (μ) and a neutrino (ν_μ) via the following reaction. The neutrino has zero rest mass.

$$\pi^+ \rightarrow \mu^+ \nu_\mu.$$

- (a) Using the relationship between total relativistic energy and momentum, show that the magnitude of the momentum p of the decay products in the pion rest frame is given by $p/c = (M_0^2 - m_0^2)/2M_0$; where M_0 is the pion rest mass and m_0 is the muon rest mass.
- (b) Consider the decay life time to be 2.5×10^{-8} sec in the rest frame of pion. Assuming that the π meson has kinetic energy equal to its rest energy, what distance would the meson travel before decaying as seen by an observer at rest?

- (a) Total momentum is conserved. Hence, the decay products will have momentum equal in magnitude, p . the Energy will be conserved. So, we have $E_\pi = E_\mu + E_\nu$. In the π rest frame:

$$\begin{aligned}
 (E_\mu + E_\nu)^2 &= E_\pi^2 \\
 \implies E_\mu^2 + E_\nu^2 + 2E_\mu E_\nu &= M_0^2 c^4 \\
 \implies (p^2 c^2 + m_0^2 c^4) + (p^2 c^2) + 2pc\sqrt{p^2 c^2 + m_0^2 c^4} &= M_0^2 c^4 \\
 \implies 2(p/c) \left[p/c + \sqrt{(p/c)^2 + m_0^2} \right] &= M_0^2 - m_0^2.
 \end{aligned}$$

Setting $p/c = w > 0$, $w^2 + \sqrt{w^4 + m_0^2 w^2} = (M_0^2 - m_0^2)/2 \implies w = p/c = (M_0^2 - m_0^2)/2M_0$.

- (b) We have $K = E_\pi \implies (\gamma - 1)M_0 c^2 = M_0 c^2 \implies \gamma = 2 \implies v = \frac{\sqrt{3}}{2}c$. So,

$$\begin{aligned}
 \text{distance in ground frame} &= (\text{speed}) \times (\text{time in ground frame}) \\
 &= \left(\frac{\sqrt{3}}{2}c \times 2 \times 2.5 \times 10^{-8} \right) \text{ m} \\
 &\approx 12.99 \text{ m.}
 \end{aligned}$$

3 | Tutorial 3

Try this final set of tutorial problems. I hope the clarity would've been better this time! 😊
Click on [Problem x.](#) to go to the solution to problem x.

3.1 • Problems

Problem 1. A particle of rest mass M having energy E (moving along x axis) decays into two particles of equal rest mass. In the lab frame, one of them is emitted along $+y$ axis. Consider the problem in x, y plane only. What are the energies of the created particles? Check your result with four momentum conservation also.

Problem 2. Consider the following one-dimensional situation, where all the motion is along the x axis. A particle has energy E' and momentum p' in frame S' . Frame S' moves at speed v with respect to frame S , in the positive $x = x'$ direction. What are E and p in S ?

Problem 3. Derive the relativistic acceleration transformation

$$a'_x = \frac{a_x \left[1 - \frac{v^2}{c^2}\right]^{3/2}}{\left[1 - u_x v/c^2\right]^3}$$

where $a_x = du_x/dt$ and $a'_x = du'_x/dt'$.

Problem 4. The photoelectric effect suggests that light of frequency ν can be regarded as consisting of photons of energy $E = h\nu$, where $h = 6.63 \times 10^{-27}$ erg-sec.

- (a) Visible light has a wavelength in the range of 400 – 700 nm. What are the energy and frequency of a photon of visible light?
- (b) A standard microwave in our kitchen operates at roughly 2.5 GHz with a max power of 7.5×10^9 erg/sec. How many photons per second can it emit?

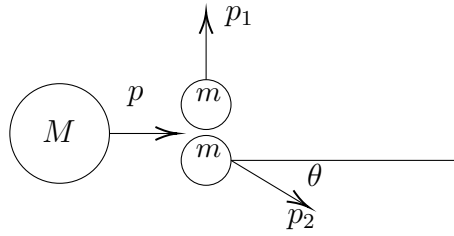
Problem 5. In an experiment performed by Jonsson in 1961, electrons were accelerated through a 50 kV potential towards two slits separated by a distance d in order to perform double-slit experiment with electrons. Calculate the electron's de Broglie wavelength, λ .

3.2 • Solutions

Click on [Solution to Problem x.](#) to go back to problem x.

Solution to Problem 1.

A particle of rest mass M having energy E (moving along x axis) decays into two particles of equal rest mass. In the lab frame, one of them is emitted along $+y$ axis. Consider the problem in x, y plane only. What are the energies of the created particles? Check your result with four momentum conservation also.



Let E be the energy of particle of rest mass M with momentum p . Let E_1, E_2 be the energy corresponding to p_1, p_2 respectively. By conservation of momentum, we have that

$$p_2 \cos \theta = p \quad \text{and} \quad p_2 \sin \theta = p_1 \\ \implies p^2 = p_2^2 - p_1^2.$$

By definition, $E_1^2 = p_1^2 c^2 + m^2 c^4$ and $E_2^2 = p_2^2 c^2 + m^2 c^4$ which implies that $E_2^2 - E_1^2 = (p_2^2 - p_1^2) c^2 = p^2 c^2 = E^2 - M^2 c^4 \implies E^2 = E_2^2 - E_1^2 + M^2 c^4$.

By Energy conservation, $E = E_1 + E_2$ and it follows that

$$(E_1 + E_2)^2 = E^2 \\ \implies E_1^2 + E_2^2 + 2E_1 E_2 = E_2^2 - E_1^2 + M^2 c^4 \\ \implies 2E_1(E_1 + E_2) = M^2 c^4 \\ \implies E_1 = \frac{M^2 c^4}{2E}.$$

And, finally, $E_2 = E - E_1 = E - \frac{M^2 c^4}{2E}$.

Note that $m \neq M/2$. It might just be a common misconception.

Solution to Problem 2.

Consider the following one-dimensional situation, where all the motion is along the x axis. A particle has energy E' and momentum p' in frame S' . Frame S' moves at speed v with respect to frame S , in the positive $x = x'$ direction. What are E and p in S ?

Let the particle move with velocity dx/dt . Observe that (if $\gamma_v = 1/\sqrt{1 - v^2/c^2}$),

$$\begin{aligned} E &= \frac{m_0}{\sqrt{1 - \frac{1}{c^2} \cdot (dx/dt)^2}} c^2 \\ &= \frac{m_0 c^3}{\sqrt{c^2 (dt)^2 - (dx)^2}} dt \\ &= \left(\frac{m_0 c^3}{dS} \right) \cdot dt, \end{aligned}$$

where dS is the invariant interval. Similarly,

$$\begin{aligned} p &= \frac{m_0}{\sqrt{1 - \frac{1}{c^2} \cdot (dx/dt)^2}} \frac{dx}{dt} \\ &= \left(\frac{m_0 c}{dS} \right) \cdot dx. \end{aligned}$$

Therefore, $E' = \left(\frac{m_0 c^3}{dS} \right) \cdot dt'$ and $p' = \left(\frac{m_0 c}{dS} \right) \cdot dx'$. By Inverse Lorentz transformations,

$$\begin{aligned} E &= \left(\frac{m_0 c^3}{dS} \right) \cdot \gamma_v \left(dt' + \frac{v}{c^2} dx' \right) \\ \implies E &= \gamma_v (E' + v p') \end{aligned}$$

$$\begin{aligned} p &= \left(\frac{m_0 c}{dS} \right) \cdot \gamma_v (dx' + v dt') \\ \implies p &= \gamma_v \left(p' + \frac{v}{c^2} E' \right). \end{aligned}$$

Invariant intervals make solutions neat, elegant and simple. We'd have to do a spamming of γ s had we not put the invariant interval to proper use.

Solution to Problem 3.

Derive the relativistic acceleration transformation

$$a'_x = \frac{a_x \left[1 - \frac{v^2}{c^2} \right]^{3/2}}{\left[1 - u_x v / c^2 \right]^3}$$

where $a_x = du_x/dt$ and $a'_x = du'_x/dt'$.

$$\begin{aligned}
a'_x &= \frac{du'_x}{dt'} = \frac{du'_x/dt}{dt'/dt} \\
&= \frac{\frac{a_x(1-\frac{u_x v}{c^2}) - (\frac{-v}{c^2} a_x)(u_x - v)}{(1-\frac{u_x v}{c^2})^2}}{\gamma(1-\frac{u_x v}{c^2})} \\
&= \frac{a_x(1-\frac{v^2}{c^2})}{\gamma(1-\frac{u_x v}{c^2})^3} = \frac{a_x}{\gamma^3(1-\frac{u_x v}{c^2})^3} \\
&= \frac{a_x \left[1 - \frac{v^2}{c^2}\right]^{3/2}}{[1 - u_x v/c^2]^3}.
\end{aligned}$$

Solution to Problem 4.

The photoelectric effect suggests that light of frequency ν can be regarded as consisting of photons of energy $E = h\nu$, where $h = 6.63 \times 10^{-27}$ erg-sec.

- Visible light has a wavelength in the range of 400 – 700 nm. What are the energy and frequency of a photon of visible light?
- A standard microwave in our kitchen operates at roughly 2.5 GHz with a max power of 7.5×10^9 erg/sec. How many photons per second can it emit?

(a)

$$\begin{aligned}
4 \times 10^{-7} &\leq \lambda \text{ (in m)} \leq 7 \times 10^{-7} \\
4.285 \times 10^{14} \text{ Hz} &= \frac{3}{7} \times 10^{15} \leq \nu = \frac{c}{\lambda} \text{ (in Hz)} \leq \frac{3}{4} \times 10^{15} = 7.5 \times 10^{14} \text{ Hz} \\
2.841 \times 10^{-12} \text{ erg} &\leq E = h\nu \leq 4.972 \times 10^{-12} \text{ erg}.
\end{aligned}$$

- Frequency $\nu = 2.5 \times 10^9$ GHz. So, energy of one photon equals $h\nu \approx 16.575 \times 10^{-18}$ erg. So, max. number of photons emitted per second equals

$$\frac{7.5 \times 10^9}{16.575 \times 10^{-18}} \approx 4.52 \times 10^{26}.$$

Solution to Problem 5.

In an experiment performed by Jonsson in 1961, electrons were accelerated through a 50 kV potential towards two slits separated by a distance d in order to perform double-slit experiment with electrons. Calculate electron's de Broglie wavelength, λ .

For a photo-electron,

$$\lambda_{\text{de Broglie}} = \frac{h}{p}.$$

Now $\frac{p^2}{2m_e} = e\Delta V \implies p = \sqrt{2m_e e\Delta V}$. So, $\lambda = \frac{h}{\sqrt{2m_e e\Delta V}} \approx 5.49 \times 10^{-12} \text{ m}$.